# MOHAMED GOMAA MOHAMED


DATA ANALYST | BUSINESS INTELLIGENCE & DATABASE DEVELOPMENT SPECIALIST

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## PROFESSIONAL SUMMARY


Analytical and business-oriented **Data Analyst** with a background in Software Engineering (AASTMT) and strong expertise in **SQL Server, Power BI, DAX, and Database Design**. Proven track record in structuring relational models (**Star Schema, 3NF**), engineering automated ETL pipelines, and delivering executive dashboards across large-scale datasets (**$20M+ revenue, 40M+ attendees**). Skilled in applying advanced SQL analytics (Window Functions, CTEs) and statistical diagnosis to eliminate operational bottlenecks and optimize business KPIs.


## TECHNICAL SKILLS

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|---|---|
| **Databases & SQL:** | Microsoft SQL Server (SSMS), SQL Querying, Relational Database Design & Normalization (3NF), Advanced Window Functions (LAG, DENSE_RANK, ROW_NUMBER), Complex Joins, CTEs, Aggregations, DML/DDL. |
| **Business Intelligence:** | Microsoft Power BI Desktop & Service, DAX (Time Intelligence, CALCULATE, Iterators), Data Modeling (Star Schema), Interactive Dashboard Architecture, KPI Scorecards, Executive Reporting, Drill-through. |
| **Data Analysis & ETL:** | Advanced Microsoft Excel (Pivot Tables, Power Pivot, XLOOKUP, Nested Logic), Power Query (M Code), Data Cleaning & Cleansing, Exploratory Data Analysis (EDA), Statistical Aggregation. |
| **Programming & Tools:** | Python (Pandas, NumPy, Matplotlib, Seaborn), Jupyter Notebook, Git & GitHub. |
| **Analytical Competencies:** | Operational Bottleneck Discovery, SLA Compliance Analysis, Customer Retention & Churn Analysis, Profit Margin Optimization, Omnichannel Analytics, Geospatial Mapping, Data Storytelling. |

## PRACTICAL DATA & BI PROJECTS

### Hajj Operations Intelligence & Bottleneck Discovery

SQL Server (SSMS) | Window Functions (LAG, DENSE_RANK) | SLA Analytics

- **Problem:** Complex pilgrimage logistics suffered from unmonitored transit delays, service station wait-time spikes, and emergency response SLA breaches across thousands of operational records.
- **Solution:** Architected a multi-table relational schema (Journeys, Routes, Services, Camps, Incidents) in **SQL Server**; formulated advanced queries utilizing `LAG()`, `DENSE_RANK()`, CTEs, and conditional grouping.
- **Result:** Isolated two primary corridors (Muzdalifah-Mina & Mina-Arafat) driving **81.5% of major transit delays (>15 mins)** and **16,100+ delay minutes**; identified Day 3 delay peaks (+1.47 min DoD) and ranked critical service bottlenecks (Water: 10.31 mins avg wait).

### FIFA World Cup Historical Analytics & Executive Dashboard

Power BI | DAX | Python (Pandas) | Star Schema

- **Problem:** Fragmented historical tournament logs spanning 22 editions lacked an audited, high-performance data model to analyze match dynamics, attendance expansion, and team conversion rates.
- **Solution:** Built an automated data cleaning and ingestion pipeline with **Python (Pandas)**; engineered a normalized **Star Schema** (Facts: match/team logs; Dimensions: tournaments, stadiums, teams) with strict 1-to-many (1:*) unidirectional filter propagation.
- **Result:** Deployed an interactive 5-page **Power BI** dashboard auditing **964 matches, 2,720 goals, and 40M+ cumulative attendees**; engineered custom DAX measures evaluating host-nation advantage boost and scoring distributions.

### Global Importers — Enterprise Sales & Profitability Dashboard

Power BI | Power Query (ETL) | DAX | Margin Analysis

- **Problem:** Leadership faced critical visibility blind spots in distinguishing high-margin revenue drivers from operational cost drains across large-scale commercial sales.
- **Solution:** Developed an automated ETL pipeline in **Power Query** to cleanse and integrate transactional records; established a relational model and formulated dynamic DAX measures for net profitability and cost analysis.
- **Result:** Delivered executive visibility over **$20M+ gross revenue** and **1M+ units sold**, verifying a **49.92% profit margin** against **$9.96M costs**; identified top-performing margin drivers (Road Bikes) and ranked high-revenue states.

### Amazon Customer Analytics & Retention Intelligence Dashboard

Power BI | DAX | Star Schema | Customer Churn

- **Problem:** E-commerce leadership lacked granular visibility into customer retention velocity, one-time buyer drop-offs, and category-level customer churn patterns.
- **Solution:** Designed a dimensional **Star Schema** in **Power BI** and formulated DAX measures tracking customer lifecycle metrics, active/inactive statuses, and purchasing cohorts with dynamic cross-filtering.
- **Result:** Quantified customer loyalty showing a **77.13% repeat buyer rate** vs. 22.87% one-time buyers, and isolated **2,000+ churned customer accounts** across product categories to enable targeted re-engagement campaigns.

## EDUCATION

**Bachelor of Software Engineering** — 2022 – 2026 (Expected)

Arab Academy for Science, Technology & Maritime Transport (AASTMT)

## PROFESSIONAL TRAINING & CERTIFICATIONS

**Data Analysis Professional Training** — Completed

National Telecommunication Institute (NTI) — Practical, intensive training in Advanced SQL Querying, Power BI Data Modeling, ETL Workflows, and Predictive Analytics.

## LANGUAGES

**Arabic:** Native | **English:** Professional Working Proficiency